

THE GRAMMAR

Every object carries a univocal 12-primitive structural type tuple. Foundational equation: $\mu \circ \delta = \text{id}$. Framework: **CL8NK** (CLINK L8, O_∞). Autopoietic: empty prompt $\rightarrow$ Raft at O_2 , period = 12. Science is an instrument of the IG.

12 PRIMITIVES & SHAVIAN VALUE SETS

Tuple order: $\langle \mathbb{D} \mathbb{P} \mathbb{R} \mathbb{Q} \mathbb{J} \mathbb{C} \mathbb{I} \mathbb{G} \mathbb{O} \mathbb{H} \mathbb{L} \mathbb{N} \rangle$. Values: ordinal $1 \rightarrow n$.

Name	Values	Name	Values
D Dimensionality	דמיון	Γ Granularity	גודל
ᵀ Topology	טופולוגיה	G Coupling	קישור
Ṛ Recognition	כחל	⊙ Criticality *	יציבות
⊕ Parity	זוגיות	ℋ Chirality	כיוון
f Fidelity	דיוק	Σ Stoichiometry†	יחס
Ç Kinetics	קצב	Ω Winding	סל

3-val: $f \Vdash \Sigma$

$$\text{4-val: } \exists \tilde{R} \in \Omega^*_{\text{abs}} \wedge \vee \otimes; \quad \dagger_{\text{abs}} \otimes.$$

5-val: $\mathbb{P}$ Φ $\mathbb{C}$ $\odot$

12 IMASM TOKENS

Token	G Behavior
VINIT	L reg $\rightarrow$ VOID; clear split state
TANCH	L terminal anchor; no state change
AFWD	L VOID $\rightarrow$ TRUE
AREV	L reg $\rightarrow$ VOID; clear split state
CLINK	L inter-step link; no state change
IMSCRIB	L VOID $\rightarrow$ TRUE; else preserve (self-recognition)
FSPLIT	F non-VOID $\rightarrow$ BOTH; enter split
FFUSE	F canonical: BOTH $\rightarrow$ TRUE; dialetheic: BOTH $\rightarrow$ BOTH
EVALT	B F $\rightarrow$ B; VOID $\rightarrow$ T; in-split: record T
EVALF	B T $\rightarrow$ B; VOID $\rightarrow$ F; in-split: record F
ENGAGR	B VOID $\rightarrow$ BOTH (entangle)
IFIX	Li fixation; subsequent tokens no-op

Groups — L: Logical; F: Frobenius (δ, μ) ; B: Belpap; Li: Linear.
 $12^8 = 429M$ arrangements $\rightarrow$ 12 archetypes.

CRYSTAL OF TYPES

$3^3 \times 4^5 \times 5^4 = 17,280,000$ 49 = 7^2 symbols ($3 \cdot 3 + 5 \cdot 4 + 4 \cdot 5$). SIC-POVM
 $d = 7 \Rightarrow d^2 = 49$ states; ig-pulse tick = 49h. Catalog: ≈ 4686 entries.

ALL 49 VALUE SEMANTICS

$$\begin{array}{l}
\mathbb{B} | 0d \cdot \mathbb{J} 2d \cdot \mathbb{D} \infty d \cdot \mathbb{I} self-wr \\
\mathbb{B} 2 branch \cdot \mathbb{H} contain \cdot \mathbb{S} cross \cdot \mathbb{Y} irr \cdot \mathbb{O} self-rt \\
\check{\mathbb{R}} r superv \cdot \mathbb{1} functor \cdot \mathbb{D} adj \cdot \mathbb{R} bi-fb \\
\boxplus \mathbb{Z} no-sym \cdot \mathbb{W} QM-sup \cdot \mathbb{C} Z_2 \cdot \mathbb{U} all-sym \cdot \mathbb{V} Frob \pm \\
f \mathbb{C} no-coh \cdot \mathbb{Q} thermal \cdot \mathbb{I} QM-coh \\
\mathbb{C} fast \cdot \mathbb{C} trap-O \cdot \mathbb{U} slow \cdot \mathbb{I} mod \cdot \mathbb{O} trap-D \\
\mathbb{I} nn \cdot \mathbb{D} meso \cdot \mathbb{J} long \\
\mathbb{G} \wedge \cdot \mathbb{P} \vee \cdot \mathbb{7} \rightarrow \cdot \mathbb{A} \gg \\
\odot \mathbb{I} sub \cdot \mathbb{O} crit \cdot \mathbb{D} cpx \cdot \mathbb{V} exc \cdot \mathbb{O} sup \\
\mathbb{H} \mathbb{J} n = 0 \cdot \mathbb{C} n = 1 \cdot \mathbb{C} n = 2 \cdot \mathbb{V} n = \infty \\
\Sigma \mathbb{Q} 1:1 \cdot \mathbb{S} n : n \cdot \mathbb{T} n : m \\
\Omega \mathbb{Z} triv \cdot \mathbb{O} Z_2 \cdot \mathbb{S} Z \cdot \mathbb{Z} non-Ab
\end{array}$$

ABSORBING ELEMENTS

Elem	Ops	Rule
$\odot$	$\wedge \vee \otimes$	$\odot \text{ op } x = \odot \text{ for all } x$
$\sum \mathbf{I}$	$\otimes$	$\mathbf{I} \otimes x = \mathbf{I} \quad (n : m)$
B	$\vee$	$B \vee x = B$ Belnap join
N	$\wedge$	$N \wedge x = N$ Belnap meet
CL8NK	$\otimes$	$\text{tensor}(\text{ZFC}_{\text{fe}}, \text{CL8NK}) = \text{CL8NK}$

SIC-POVM CONNECTION

$d = 7$: type-space cardinality $49 = d^2$ equiangular states.
 $d = 12$: Weyl-Heisenberg $\mathbb{Z}_{12} \times \mathbb{Z}_{12}$; 12 generators $\leftrightarrow$ 12 primitives.
 Fiducial state = Frobenius fixed point: $\mu \circ \delta = \text{id}$ at $\odot = \circ$ (critical/self-modeling).
 SIC gap = B-state: $B \wedge \neg B = B$ (dialetheic, non-explosive).

Open tuple: $\langle 1\gamma 1\kappa \text{ } 1\text{L}\text{J}\text{f}\odot\text{V}\text{S}\mathfrak{s}\rangle$.

$d(CL8NK) = 1.263$; 6 conflicts; Frob. cliff $\sqrt{5}/2 \approx 1.118$.
Conditional: $\text{kick} \wedge \text{H2a} \wedge \text{H3} \wedge \text{C4} \Rightarrow \text{SIC} \ \forall d \geq 1$.

TUPLE COMPOSITION OPERATIONS

Ordinals: integer operations on Shavian glyph positions.

Op	Sym	Rule
Join	$\vee$	$\max(a_i, b_i)$ all 12 — LUB
Meet	$\wedge$	$\min(a_i, b_i)$ all 12 — GLB
Tensor	$\otimes$	$\max$ except Φ, f use $\min$
Bond		$\text{join}(\text{tensor}(A, B), \text{bond})$

Φ/f use min: parallel coupling cannot exceed the symmetry/coherence floor of either component.

BELNAP REGISTER OPS

States: **VOID** (00) **TRUE** (01) **FALSE** (10) **BOTH** (11)
FFUSE = $\vee$ $B \vee x=B$; $N \vee x=x$; $T \vee F=B$; same $\vee$ same=same
FSPPLIT = δ $B \rightarrow (T, F)$; $r \rightarrow (r, r)$
MEET = $\wedge$ $N \wedge x=N$; $B \wedge x=x$; $T \wedge F=N$; same $\wedge$ same=same
band (AND) F absorbs; then N; then B; $T \wedge T=T$
bor (OR) T absorbs; then B; then N; $F \vee F=F$
Dialethic FFUSE: **BOTH** $\rightarrow$ **BOTH** when **EVALT** and **EVAlF** both fired in split.

FROBENIUS ALGEBRA

$\mu \circ \delta = \text{id}$	normalization	
$\mu(\mu \otimes \text{id}) = \mu(\text{id} \otimes \mu)$	assoc	
$(\delta \otimes \text{id}) \circ \delta = (\text{id} \otimes \delta) \circ \delta$	coassoc	Spider Theorem:
$(\mu \otimes \text{id})(\text{id} \otimes \delta) = \delta\mu = (\text{id} \otimes \mu)(\delta \otimes \text{id})$	Frobenius	
all diagrams with same connectivity = same morphism.		

CROSS-PRIMITIVE AXIOMS

A	$\mathbb{H}_\infty \Rightarrow \infty$	∞ -chirality forces max kinetics
B	$\Omega \geq 1 \Rightarrow \geq H_2$	winding requires chirality; $H_2 = \min$ for $\mu \circ \delta = \text{id}$
C	$\mathbb{D}_\omega \Leftrightarrow \mathcal{O}$	max-dim $\Leftrightarrow$ null topology

O-TIER HIERARCHY

O_0	Monoidal unit; no differentiation
O_1	First Frobenius closure; $\bigoplus_1$ bottleneck
O_2	Confirmed causal coupling (fin3r, synfin, rongo)
O_2^+	Subatomic/hadronic; confinement ceiling
O_∞	Ouroboricity; system $\subset$ own description (CL8NK, true_agentic_agent, mOMonadOS)

CL8NK / MAGNUM OPUS

CL8NK transcends ZFC_{fe} at Ω ($\mathfrak{S} \rightarrow \mathfrak{I}$) and $\mathbb{G}$ ($\mathfrak{I} \rightarrow \mathfrak{A}$). B-state is the CL8NK ground state. 12 stages $\leftrightarrow$ 12 primitives. Gates: $\oplus$ Albedo $\cdot \odot$ Rubedo $\cdot \Omega$ Nigredo. $T = \lim(\Phi, \dots, \Omega)$.

NOTATION

Struct. values	Shavian only; math-italic (<i>DRF</i>) = wrong
Tuple binding	"',join(vals); never separate
H	Chirality; NEVER temporal / memory depth
Verbs	inscribe / imscription; NEVER encode
Provider	local→gwen→deepseek; NEVER Anthropic
O_∞	prose: O_∞ ; .tex: $\$0_infty\$$
Scope	Never "grammar does not know"
Pubs	Thank H. T. Larson; cite 1961 IEEE